

Multiple Choice

1. **Answer:** A

Options: A) By assessing the fossil content found within the rock layers; B) By comparing the density of the rock to a known age database; C) By examining the color and texture of the rock; D) By counting the layers of the rock; E) By measuring the amount of carbon-14 remaining in the rock

Note: Correct option: A - By assessing the fossil content found within the rock layers

2. **Answer:** A

Options: A) Animals use the sound of the ocean waves for directional cues in migration.; B) Animals rely solely on memory from previous migrations.; C) Animals migrate based on the patterns of the stars alone.; D) Animals follow changes in air pressure for migration.; E) Animals use temperature changes for migration.

Note: Correct option: A - Animals use the sound of the ocean waves for directional cues in migration.

3. **Answer:** A

Options: A) 108 nucleotides; B) 120 nucleotides; C) 150 nucleotides; D) 144 nucleotides; E) 60 nucleotides

Note: Correct option: A - 108 nucleotides

4. **Answer:** D

Options: A) RNase only; B) a DNA primer and RNase; C) a DNA or RNA primer and DNA polymerase; D) a DNA or RNA primer and reverse transcriptase; E) a RNA primer and DNA polymerase

Note: Correct option: D - a DNA or RNA primer and reverse transcriptase

5. **Answer:** A

Options: A) Antibiotic resistance arises from bacteria absorbing nutrients that counteract the drug; B) Resistance develops due to prolonged exposure to antibiotics; C) Bacteria develop a natural immunity to drugs over time; D) Bacteria develop resistance through spontaneous generation without exposure to antibiotics; E) Drug resistance is caused solely by the use of antibacterial hand sanitizers

Note: Correct option: A - Antibiotic resistance arises from bacteria absorbing nutrients that counteract the drug

True / False

6. **Answer:** False

Options: A) True; B) False

Note: In Southern Italy, hepatitis G virus infection is widespread among patients with

chronic hepatitis, independently of parenteral risk factors. Its frequency in subjects with cryptogenic liver disease parallels that observed in hepatitis C virus chronic liver disease, thus ruling out an aetiological role of hepatitis G virus. TT virus infection is uncommon in patients with cryptogenic or hepatitis C virus-related liver disease who do not have a history of parenteral exposure.

7. **Answer:** True

Options: A) True; B) False

Note: These findings indicate that a relationship between multiple sclerosis and streptococcal infections may exist, but to acquire a better understanding of the role of group A streptococci in the pathogenesis of multiple sclerosis, more studies with animal models are necessary.

8. **Answer:** True

Options: A) True; B) False

Note: Our study confirms that students are able to identify QA issues and propose solutions. We recommend that students have a formally recognized place in day surgery QA programmes, to close the QA loop and to adequately prepare them for medical practice in the 21 st century.

9. **Answer:** False

Options: A) True; B) False

Note: Baseline pain intensity does not predict the outcome after an appropriate opioid titration. It is likely that non-homogeneous pain treatment would have biased the outcome of a previous work.

10. **Answer:** True

Options: A) True; B) False

Note: These cases illustrate that tumours involving eyelids and parotid glands can present simultaneously or sequentially, and either of these structures could be the focus of primary or metastatic tumour. The important message for oculoplastic and parotid surgeons is to routinely assess both the periocular and parotid area when patients present with a mass in either structure.

Long Form Response

11. **Answer:** It increases diffusion of H⁺ from the intermembrane space to the matrix.. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** Fermentation. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key